

{3} Current Electricity

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- * ELECTRIC CURRENT
- * ELECTRIC CURRENTS IN CONDUCTORS
- * OHM' S LAW
- * DRIFT OF ELECTRONS AND THE ORIGIN OF RESISTIVITY
- * LIMITATIONS OF OHM 'S LAW

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* OHM'S LAW

* DRIFT OF ELECTRONS AND THE ORIGIN OF RESISTIVITY

* LIMITATIONS OF OHM'S LAW

* TEMPERATURE DEPENDENCE OF RESISTIVITY

* ELECTRICAL ENERGY, POWER

* CELLS, EMF, INTERNAL RESISTANCE

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- * **Charges in motion constitute an electric current.**
- * In the present chapter, we shall study some of the basic laws concerning **steady electric currents**.

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- * The direction of current is taken as the direction of motion +ve charges or opposite to the motion of -ve charge.

Ex 1 : In cosmic rays 0.15 protons/cm²-s are entering the earth's atmosphere. If the radius of the earth is 6400 km, what is the current received by the earth in the form of cosmic rays?

Ex 2: In neon gas discharge tube 2.9×10^{18} Ne^+ ions move to the right through a cross-section of the tube each second, while 1.2×10^{18} electrons move to the left in this time. What is the net electric current in the tube?

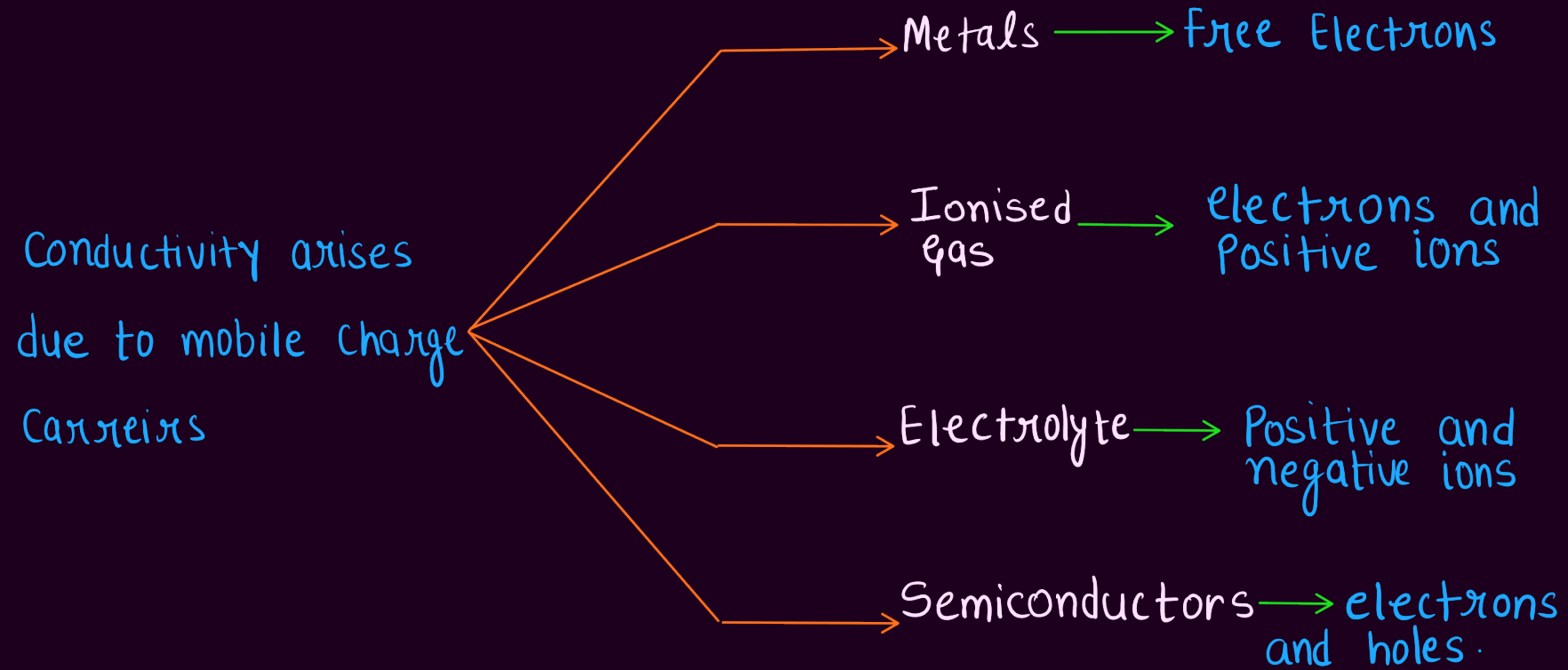
Ex 3 : In the Bohr's model of hydrogen atom, the electron is assumed to rotate in a circular orbit of radius 5×10^{-11} m, at a speed of 2.2×10^6 m/s. What is the current associated with electron motion?

Ex 4 : The current in a wire varies with time according to the relation $I = 2 \text{ A} + (0.6 \text{ A/s}^2) t^2$.

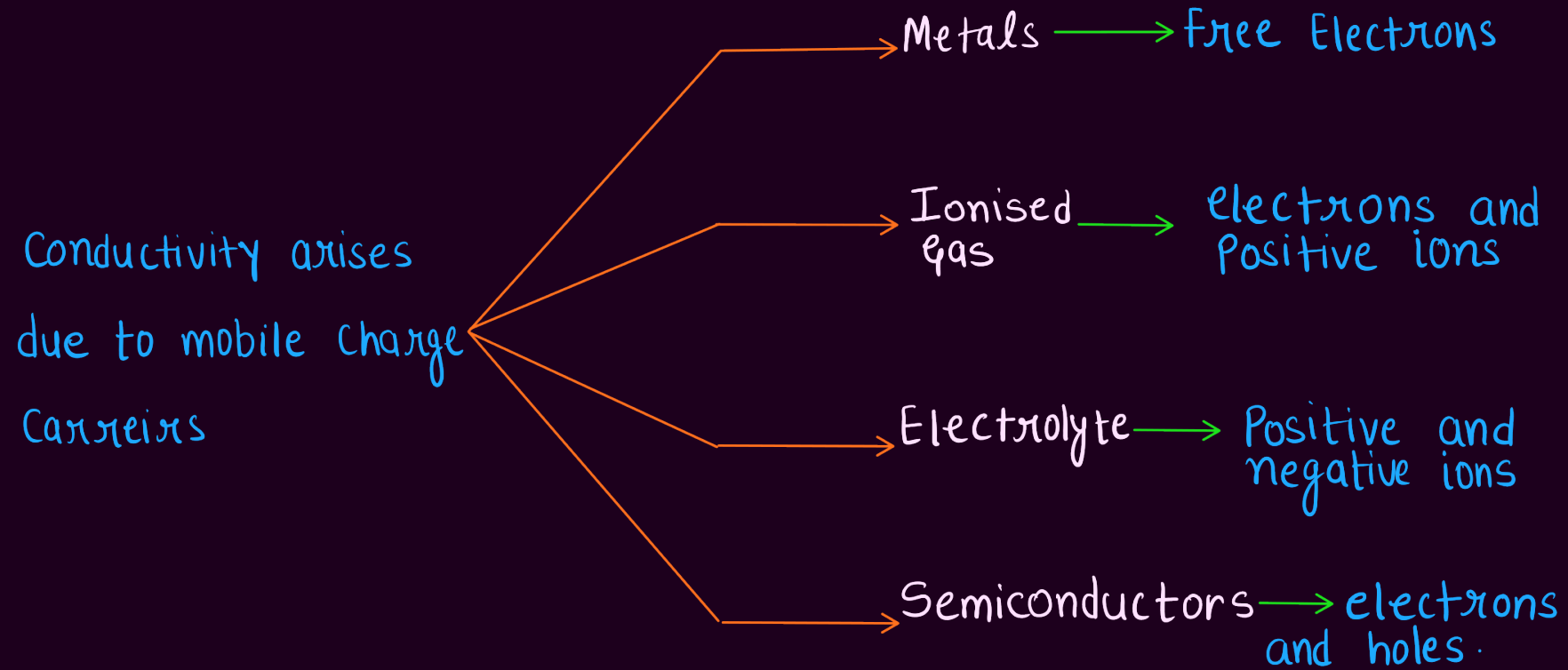
- (a) How many coulomb of charge pass a cross-section of the wire in the time interval between $t = 0$ and $t = 10 \text{ s}$?
- (b) What constant current would transport the same charge in same charge in the same time interval?

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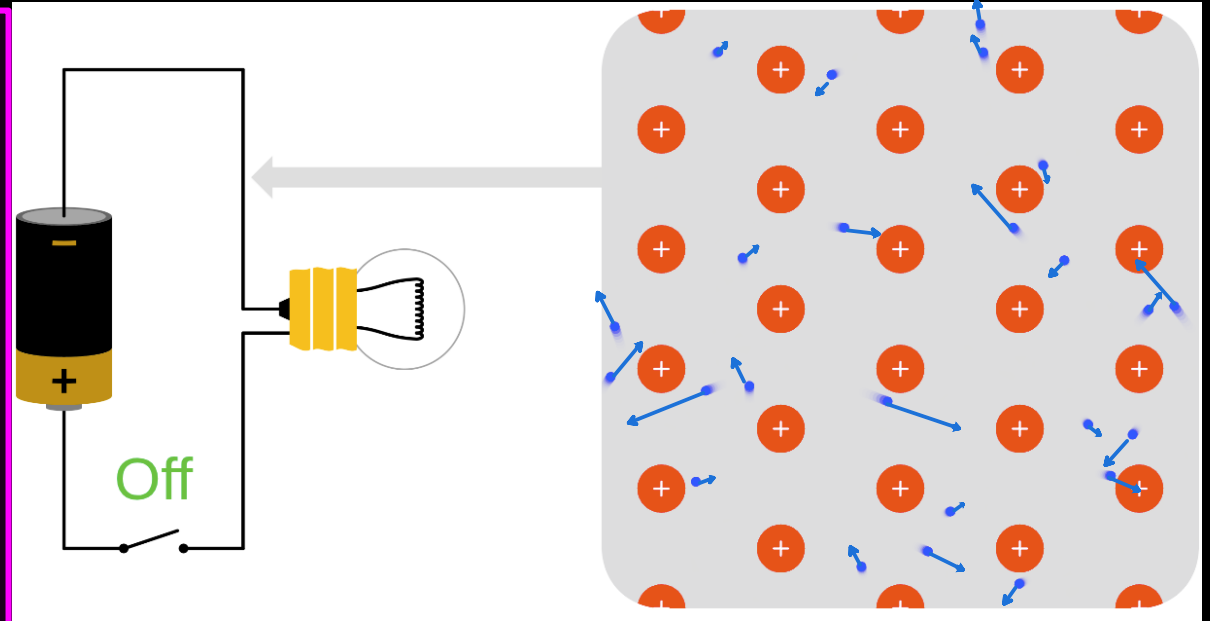


* In our discussions, we will focus only on solid conductors so that the current is carried by the negatively charged electrons in the background of fixed positive ions.

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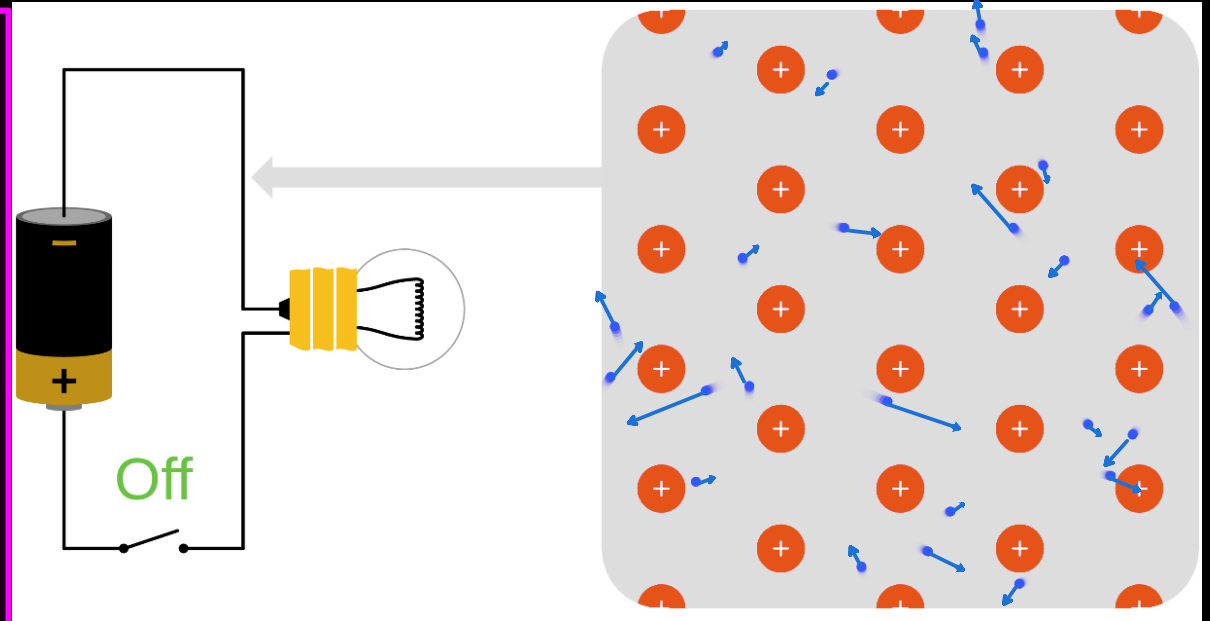
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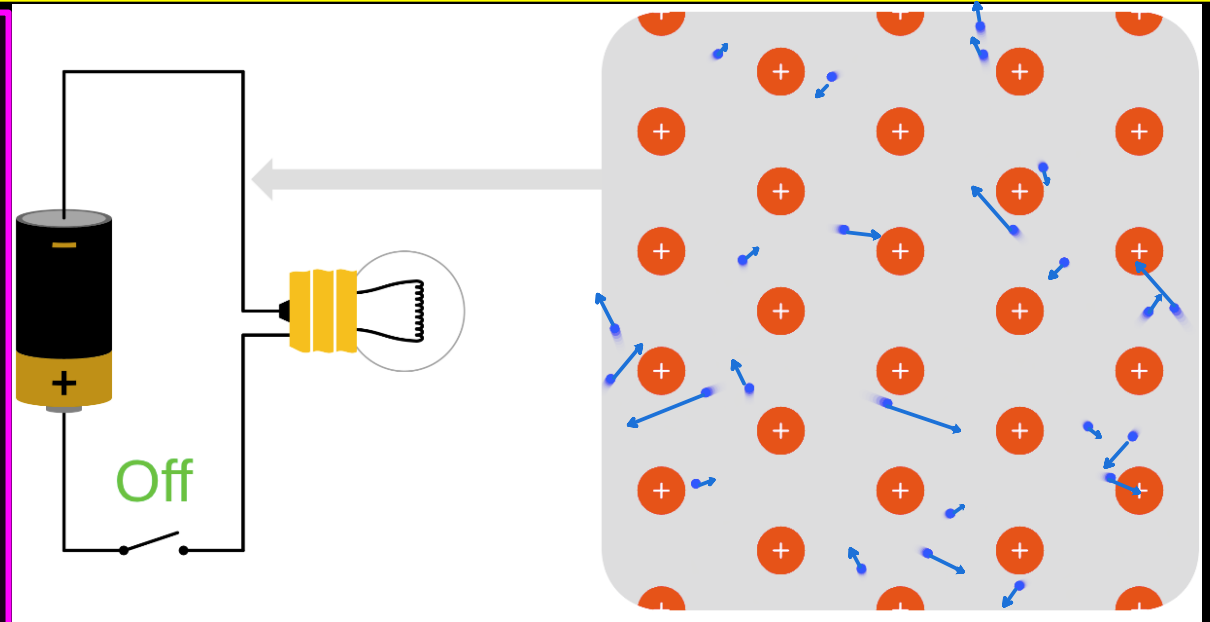
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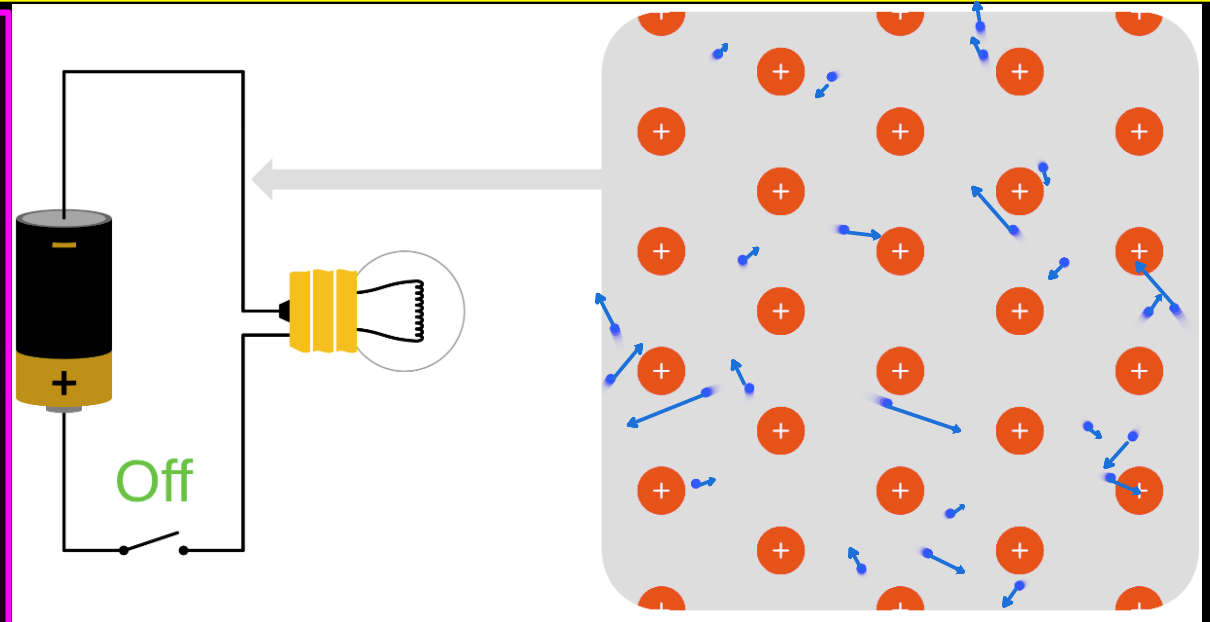


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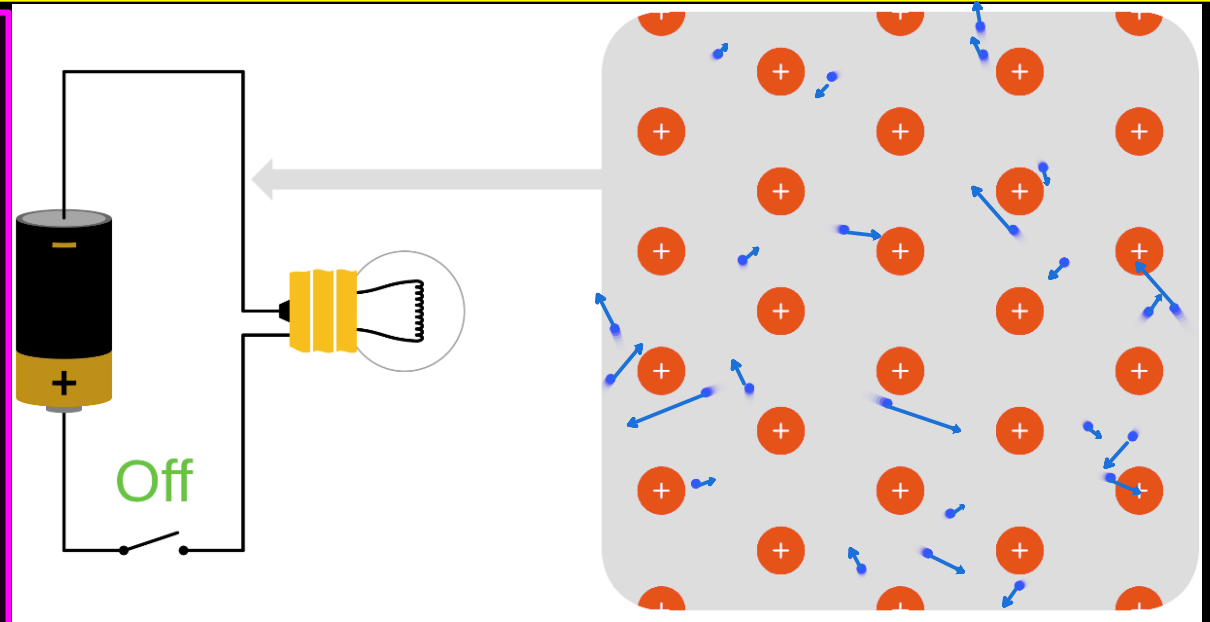


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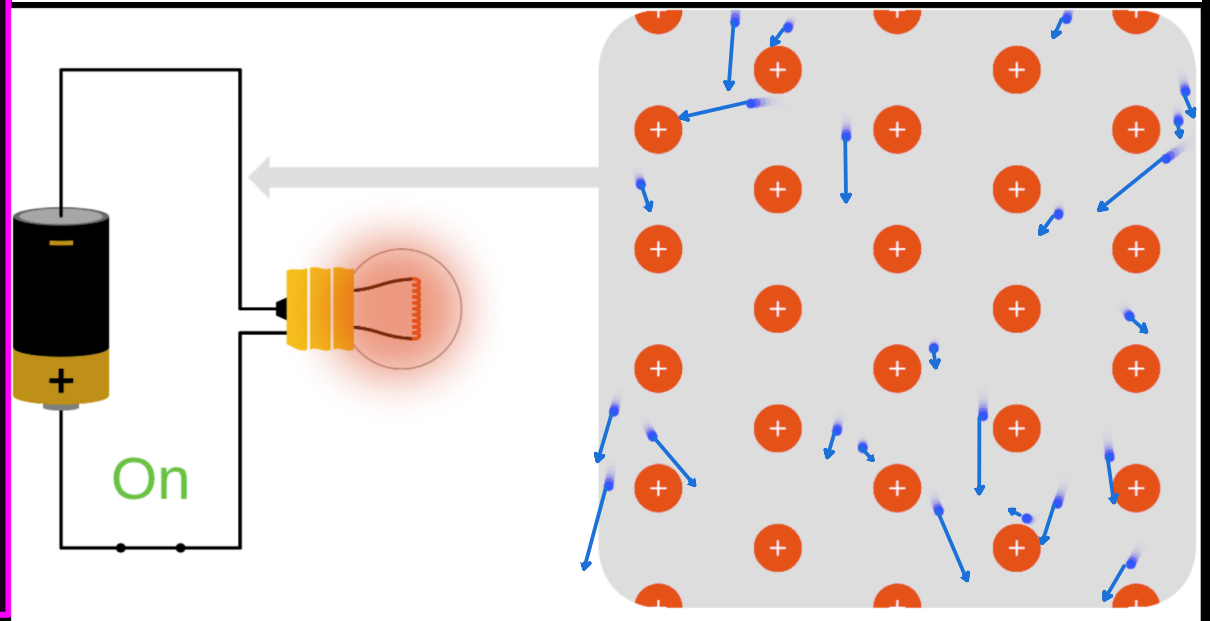
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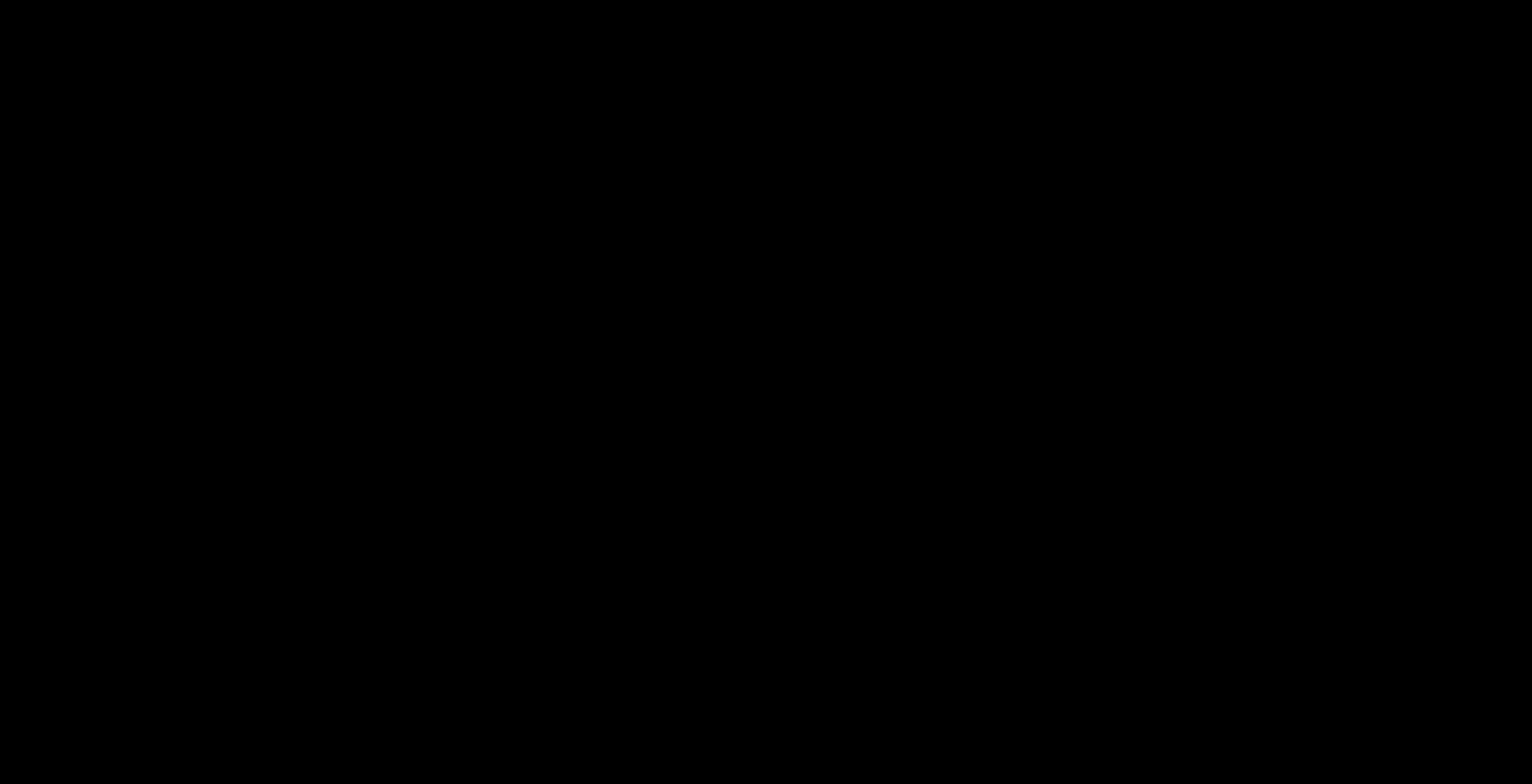


Case 2 : When electric field is present :

* The electrons will be accelerated due to this field, in the opposite direction of the field. The electrons, as long as they are moving, will constitute an electric current.



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$$R \propto \frac{l}{A} \text{ or } R = \rho \frac{l}{A}$$

* where the constant of proportionality called resistivity (ρ) depends on the material

CURRENT DENSITY AND EQUIVALENT FORM OF OHM'S LAW

Consider a rectangular block of material with length l , cross-sectional area A , and resistance R . A potential difference V is applied across the ends of the block, causing a current I to flow through it.

The current density J is defined as the current per unit cross-sectional area:

$$J = \frac{I}{A}$$

According to Ohm's Law, the current I is related to the potential difference V and the resistance R by:

$$I = \frac{V}{R}$$

Substituting the expression for I into the equation for J , we get:

$$J = \frac{V}{RA}$$

The resistance R of a rectangular block is given by:

$$R = \frac{\rho l}{A}$$

where ρ is the resistivity of the material. Substituting this expression for R into the equation for J , we get:

$$J = \frac{V}{\rho l}$$

This equation shows that the current density J is directly proportional to the potential difference V and inversely proportional to the length l of the block and the resistivity ρ of the material.

Thus, the equivalent form of Ohm's Law in terms of current density is:

$$J = \frac{V}{\rho l}$$

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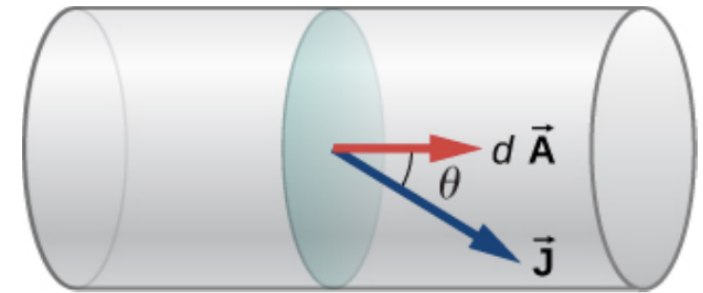


Figure 9.12 The current density $\vec{J}$ is defined as the current passing through an infinitesimal cross-sectional area divided by the area. The direction of the current density is the direction of the net flow of positive charges and the magnitude is equal to the current divided by the infinitesimal area.

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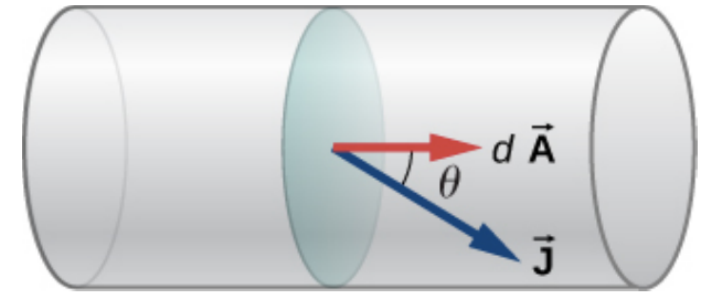


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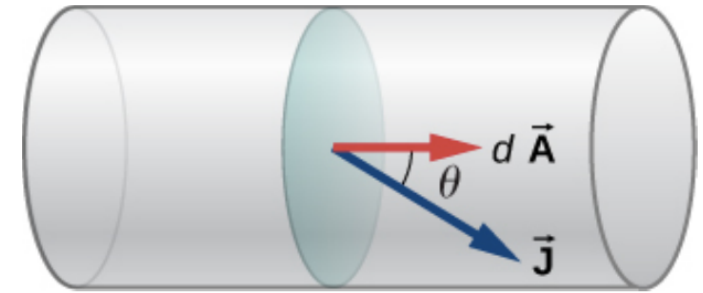


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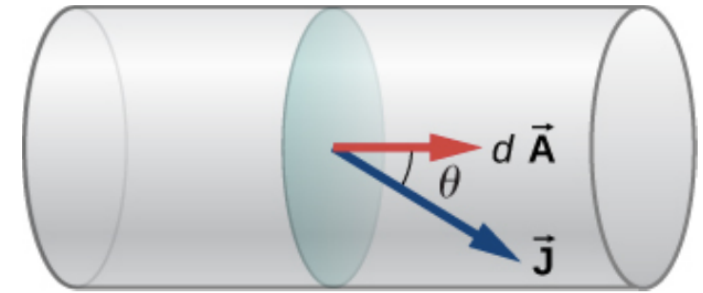


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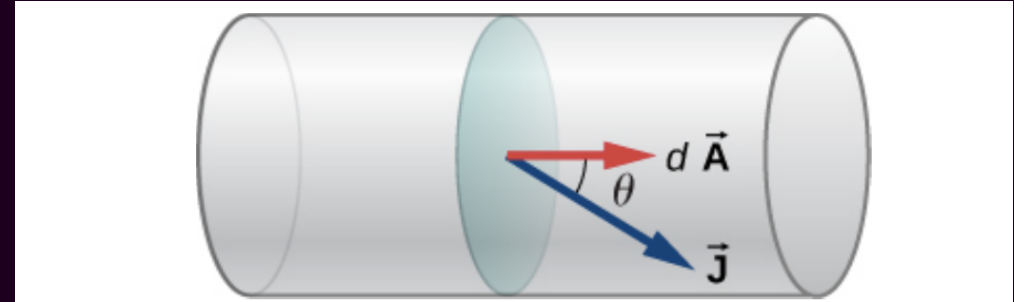


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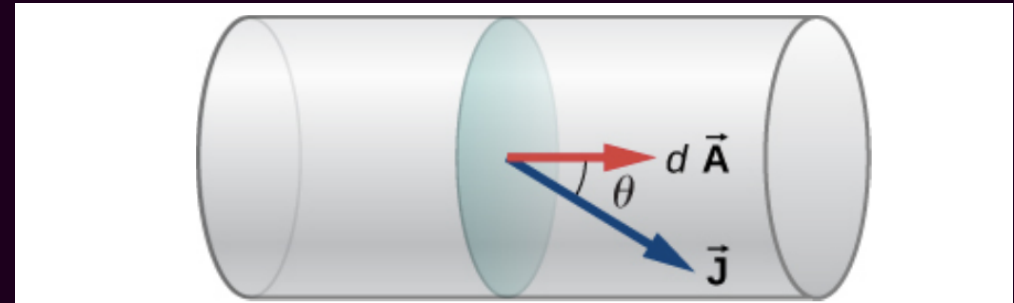


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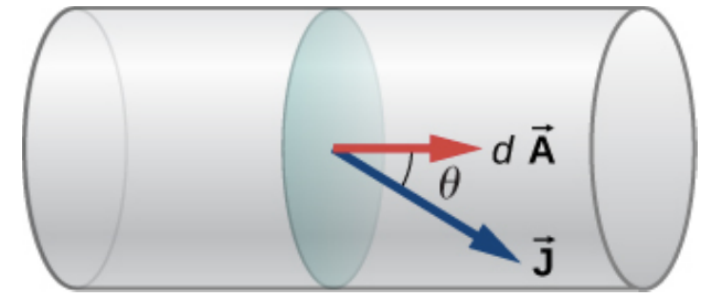
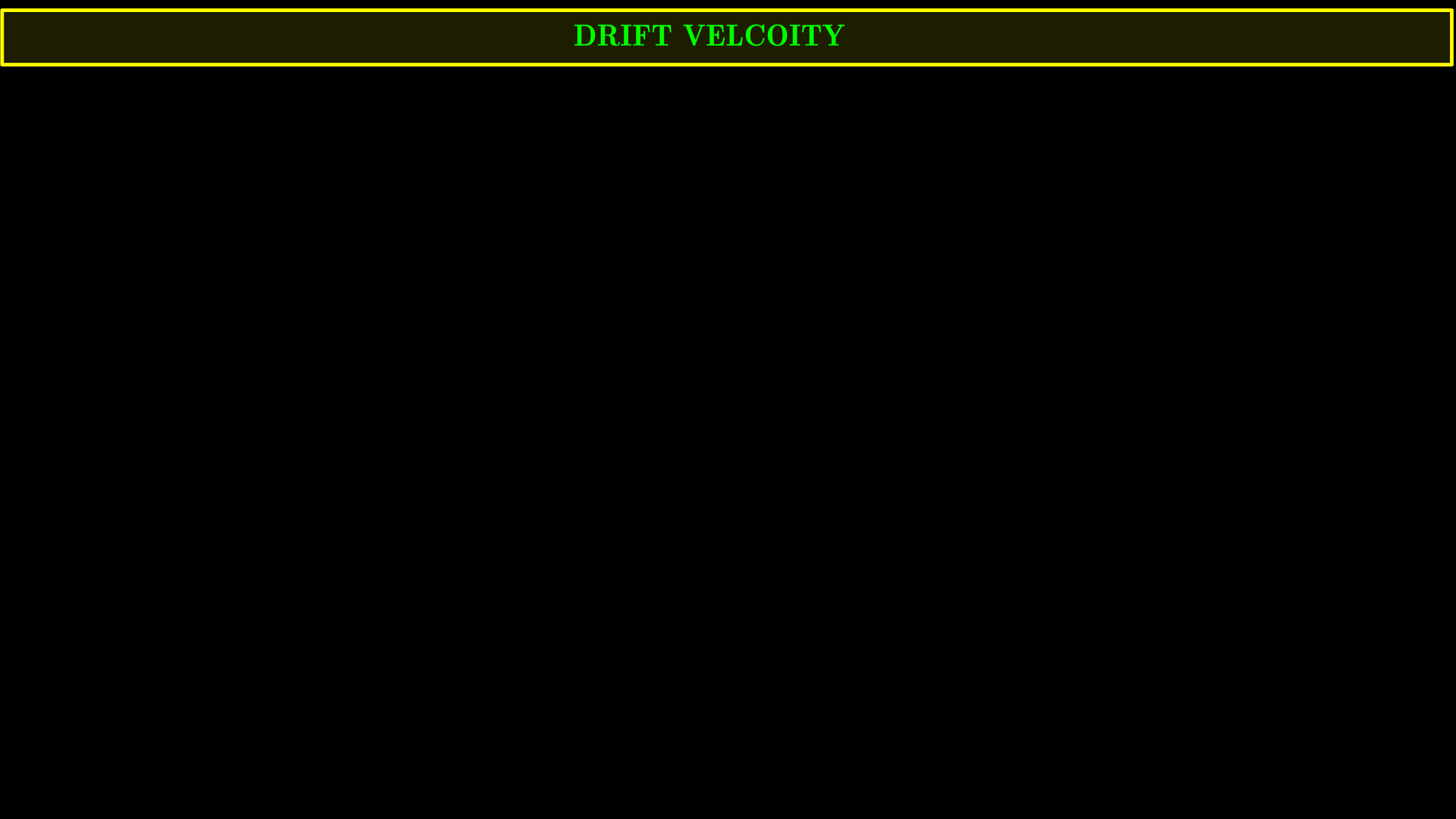


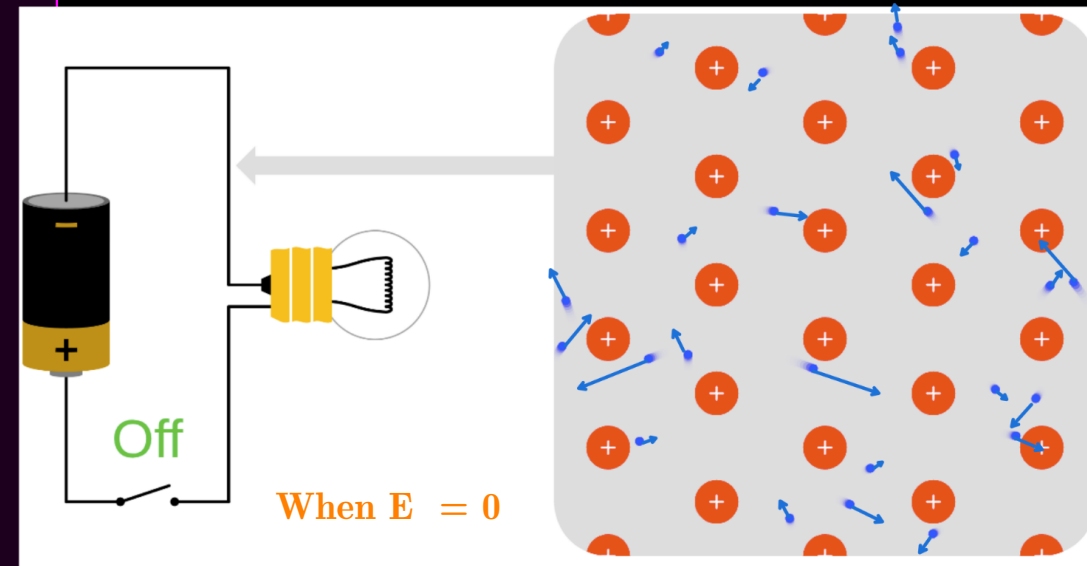
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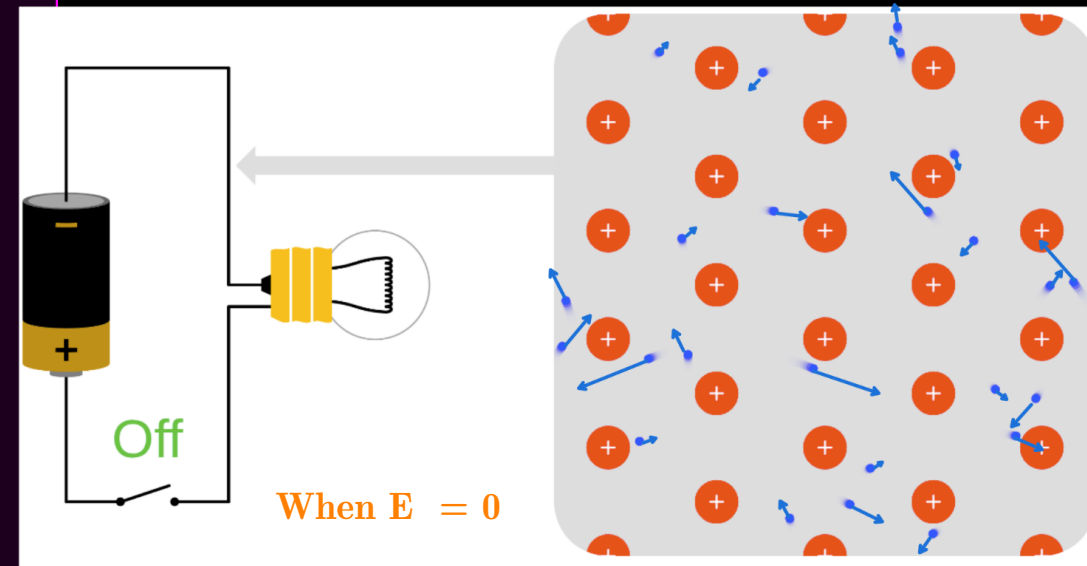
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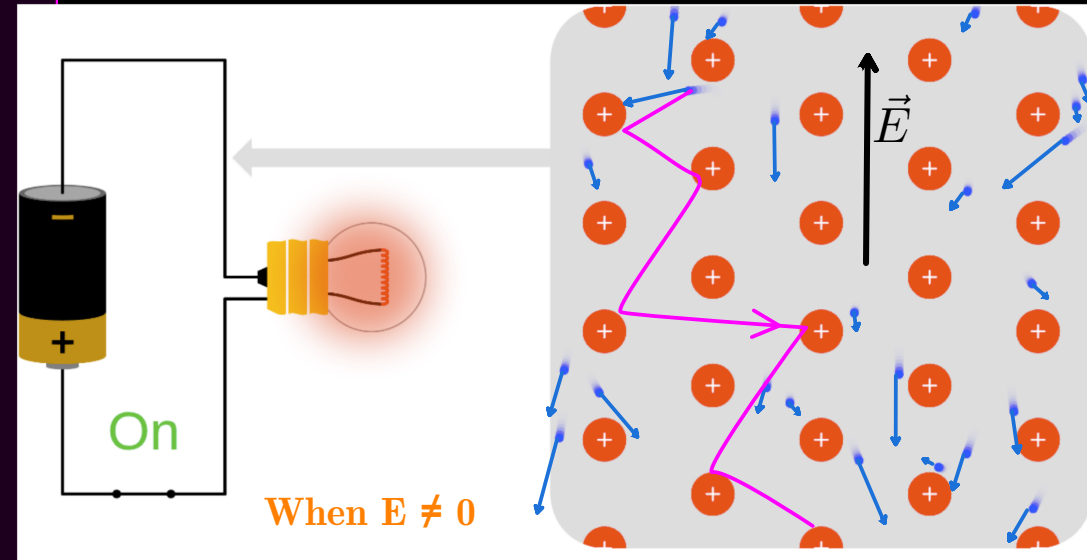
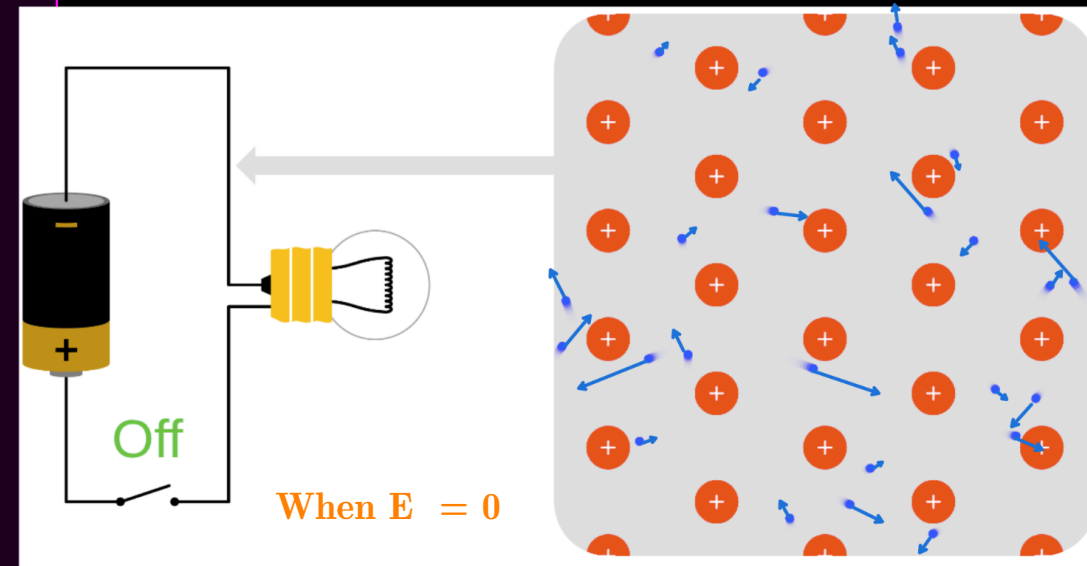
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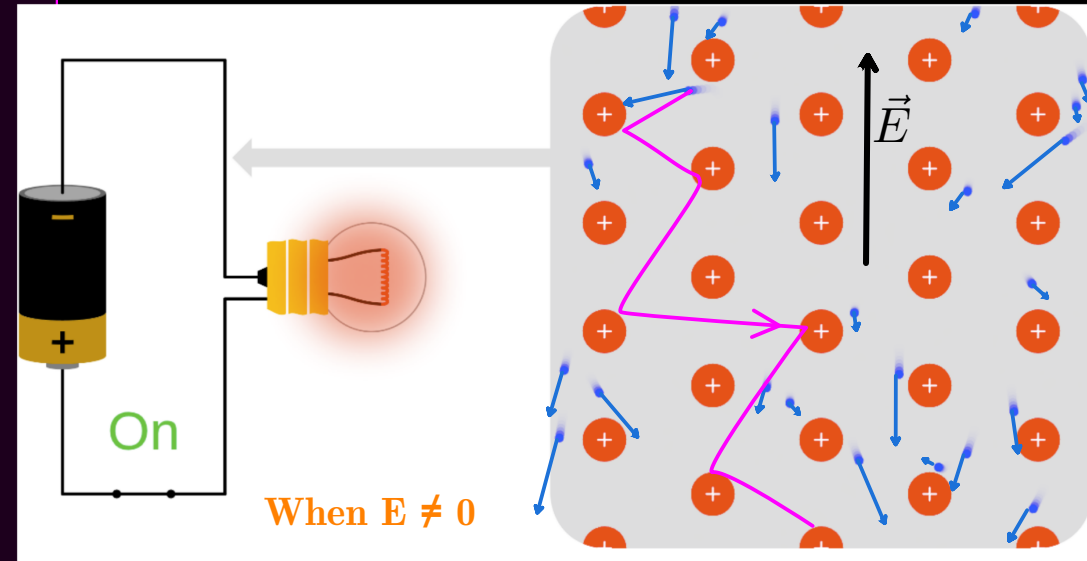
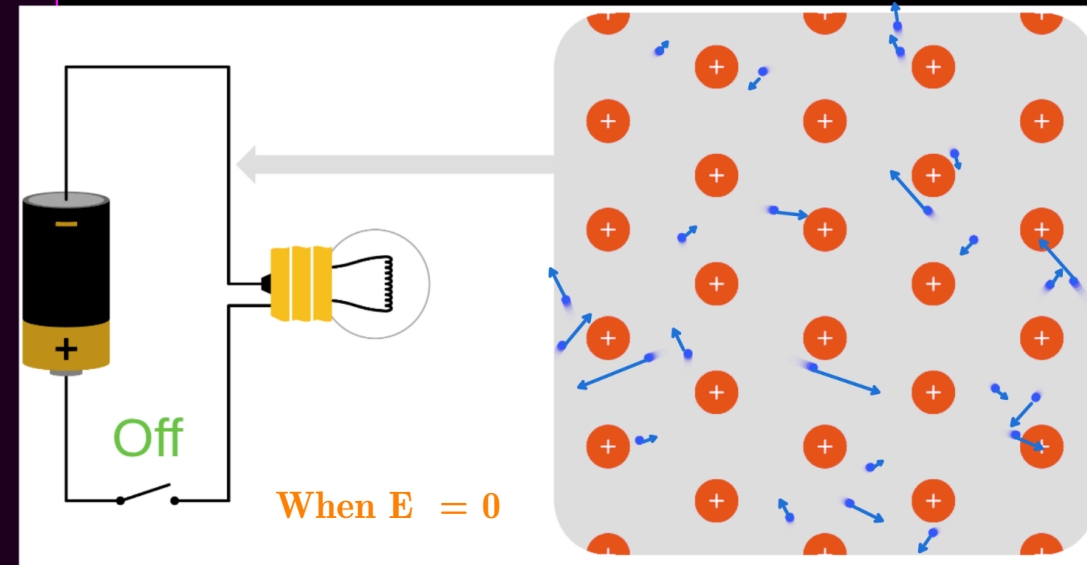
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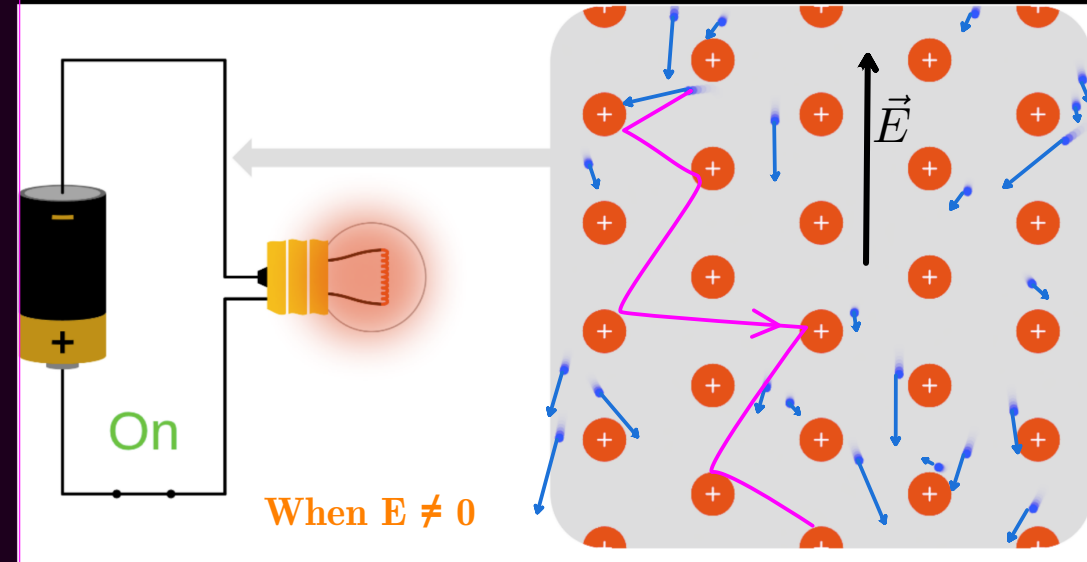
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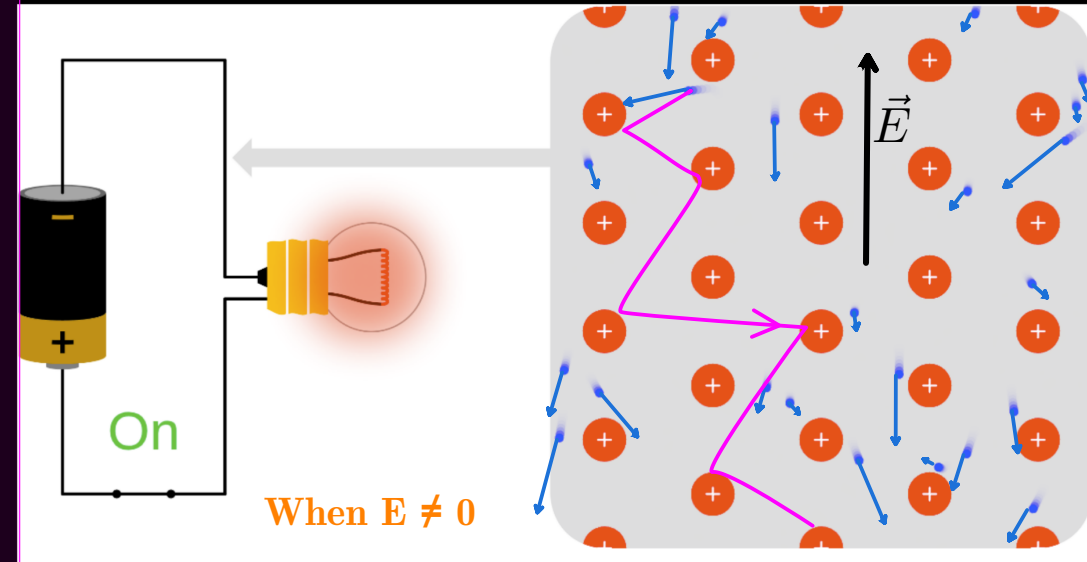
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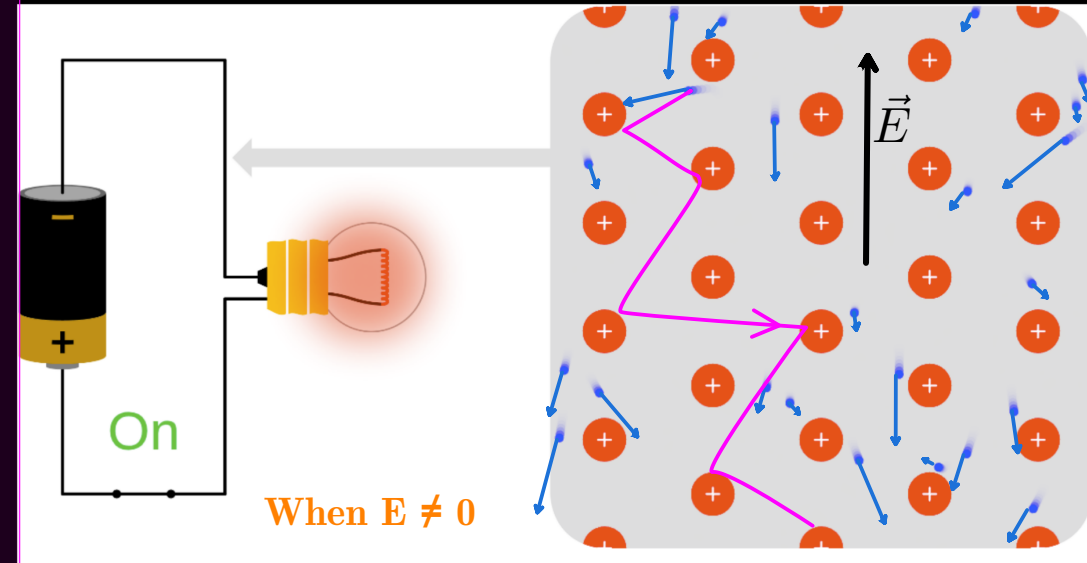


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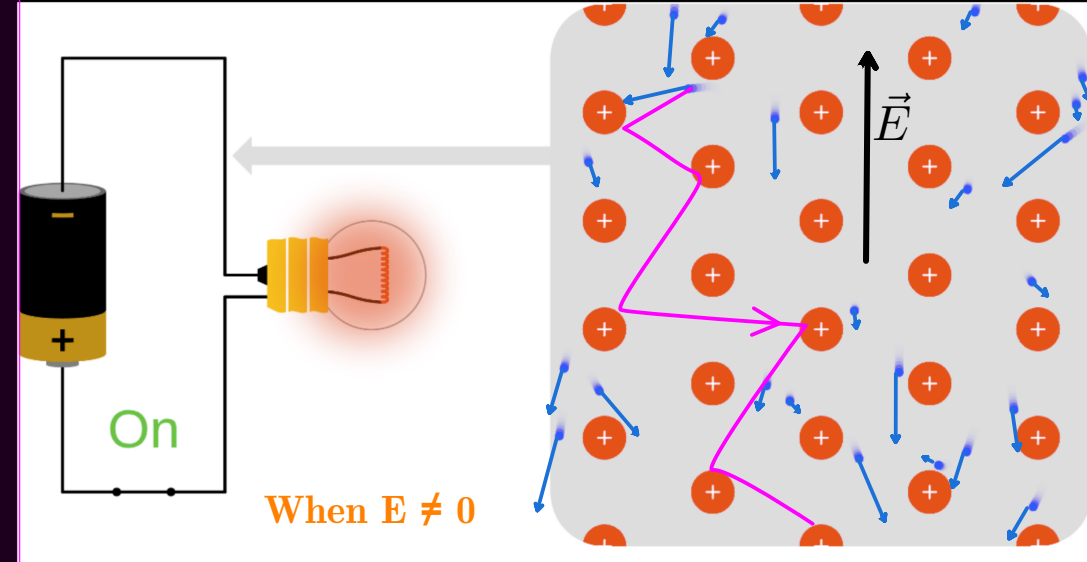
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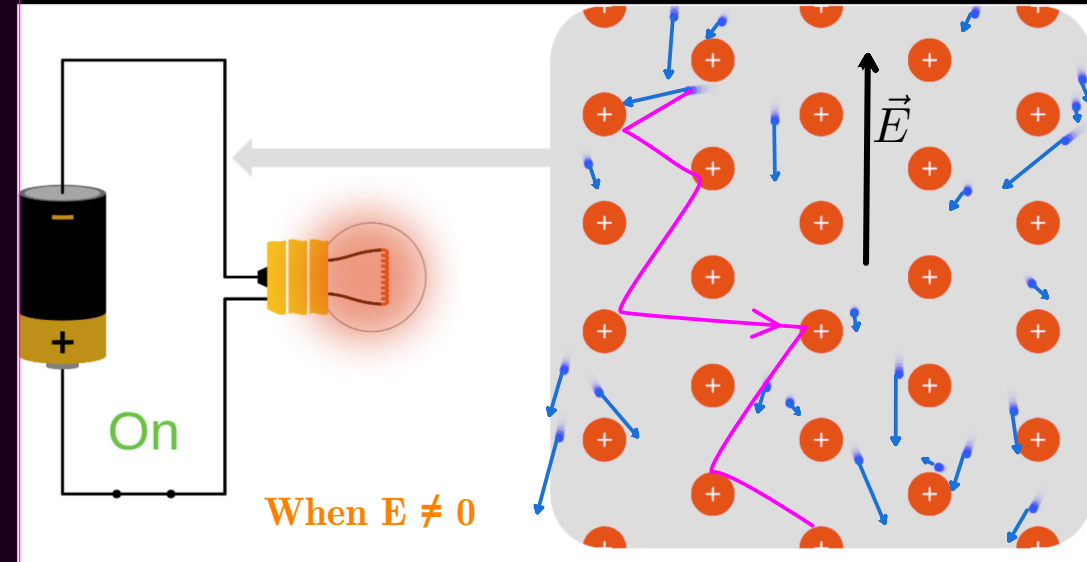
- * The average velocity gained by all n electrons (called drift velocity) will be :

$$\vec{v}_{\text{avg.}} = \frac{1}{n} \sum_{i=1}^n v_i = \frac{1}{n} \sum_{i=1}^n u_i - \frac{e\vec{E}}{m_e} \sum_{i=1}^n \frac{t_i}{n}$$

DRIFT VELOCITY

- * As the electron accelerates it gains velocity, which last for a short time and is lost in the next collision.
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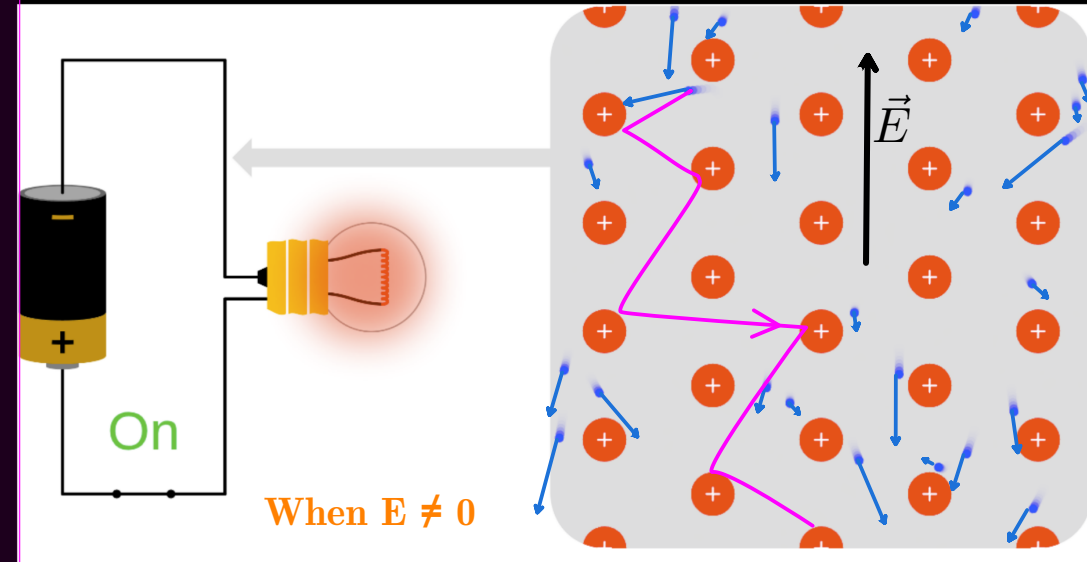
* τ is the average time between two successive collision and is known as relaxation time.

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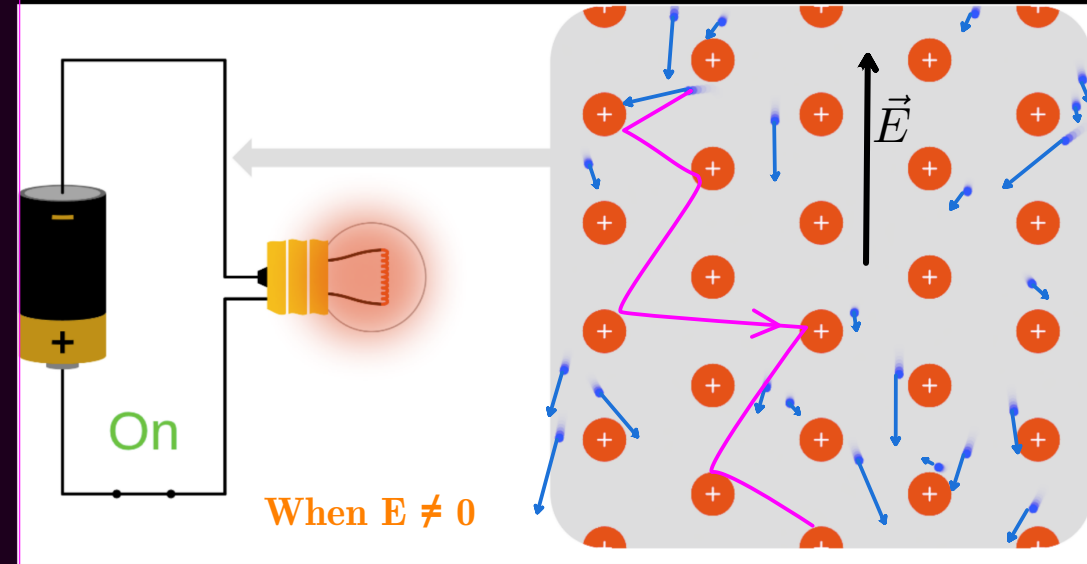
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When $E \neq 0$

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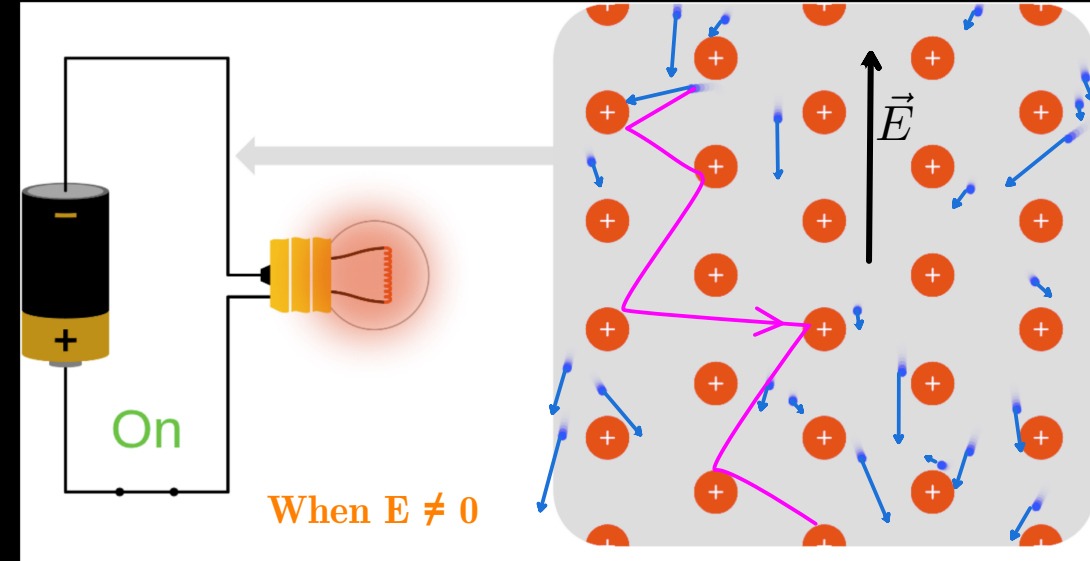
$$\vec{v}_d = 0 - e \frac{e\vec{E}}{m_e} \tau$$

$$\vec{v}_d = - \frac{e\vec{E}}{m_e} \tau$$

- * **Drift velocity :** The average velocity with which the free electrons get drifted in the opposite direction of the applied electric field.

DRIFT VELOCITY

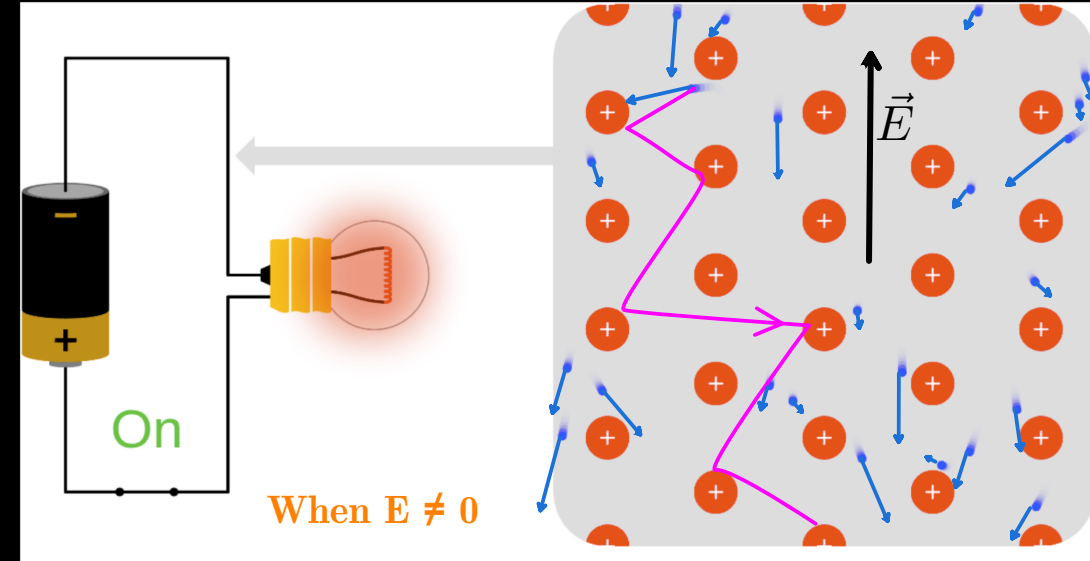
- * Due to continuous collisions of electrons and the positive ions, the speed of the electrons doesn't continuously increase. The result of this is that the electrons move within the material with some average constant velocity (drift velocity) in the direction opposite the applied electric field.



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- * The drift velocity of electrons is very small in the order of 10^{-4} ms^{-1} .

